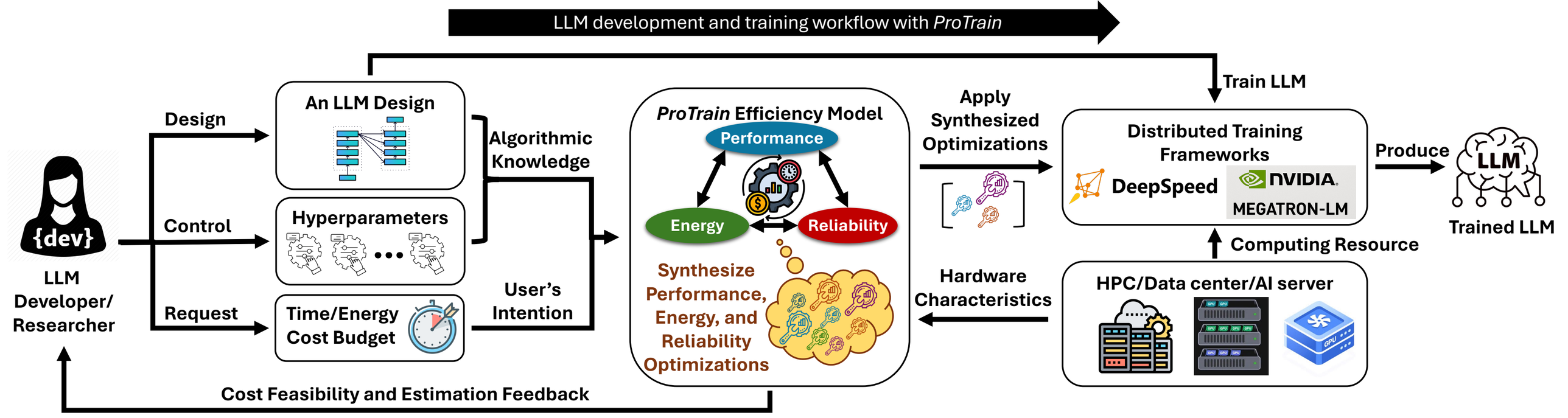
LLM development and training workflow with *ProTrain*
Train LLM
LLM Developer/ Researcher
{dev}
Design
Control
Request
An LLM Design
Hyperparameters
Time/Energy Cost Budget
Algorithmic Knowledge
User's Intention
ProTrain Efficiency Model
Performance
Energy
Reliability
Synthesize Performance, Energy, and Reliability Optimizations
Apply Synthesized Optimizations
Distributed Training Frameworks
DeepSpeed
NVIDIA
MEGATRON-LM
Produce
LLM
Trained LLM
Computing Resource
Hardware Characteristics
HPC/Data center/AI server
Cost Feasibility and Estimation Feedback